A Decision-Support Framework for Platelet Inventory Management

Demand Forecasting and Simulation-Based Comparison of Replenishment Policies

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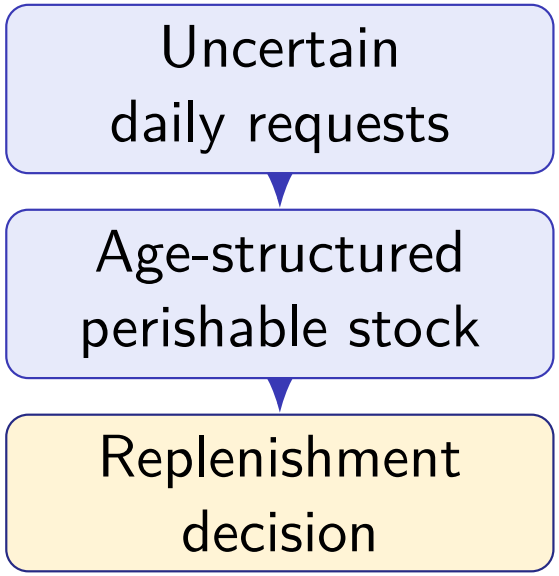
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1. Introduction — Motivation and Problem

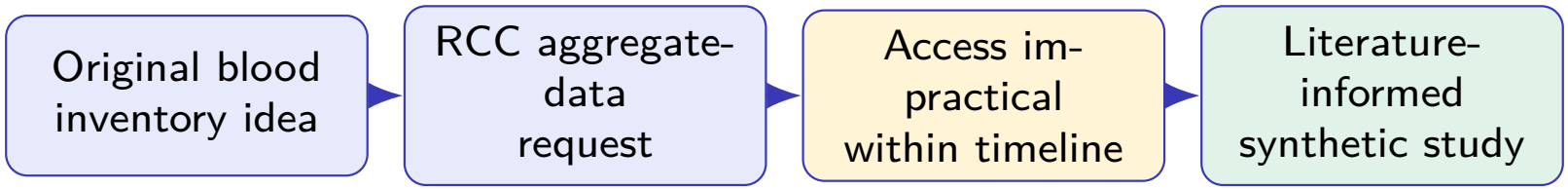
- Platelet inventory must balance uncertain demand with a short usable life.
- Excess stock may expire; insufficient eligible stock leaves demand unmet.
- Daily replenishment should consider both predicted demand and inventory age.
- We study this trade-off in a **simulation-based, single-blood-bank setting**.



Problem statement. How do established replenishment approaches compare on platelet shortages, expiry and inventory under disclosed synthetic single-bank assumptions?

General methodological motivation: Motamedi et al. (2025); Haijema et al. (2007). Findings from prior settings are not assumed to transfer directly.

1. Introduction — Project Evolution



- The team previously requested aggregate platelet data from RCC, Thiruvananthapuram.
- Suitable data were unavailable within the project timeline; published weekday statistics now inform synthetic inputs.
- No raw institutional dataset is used; published aggregate statistics inform synthetic inputs.
- Source statistics come from another setting and require local adaptation before deployment.

Published aggregate means; synthetic requests

2. Objectives and Scope

1. Generate reproducible synthetic platelet-demand inputs calibrated to published weekday transfusion means, documenting assumptions and planned sensitivity cases.
2. Compare seasonal naïve, MA7, ARIMA and weekly SARIMA chronologically, then examine their effects within a common ordering rule.
3. Implement and tune selected established replenishment approaches, comparing quantity-based and remaining-life-adjusted ordering.
4. Compare fill rate, stockout-day rate, wastage and mean inventory on matched simulated paths across independent replications and operating conditions.
5. Present conditional recommendations in a decision-support prototype.

Current scope. Single-bank pooled platelet demand; daily review; one-day lead time; uncertain receipt quantities; five remaining-life cohorts; FEFO; unmet demand without automatic backlog.

Outside scope. Other components; blood-group compatibility; patient eligibility; inter-bank sharing; clinical validation.

Completed: reference, forecast evaluation, fixed-order benchmark. Planned: policy comparisons, replications, sensitivity and prototype.

3. Literature Review (1/2) — Forecasting and Integration

Paper	Method / context	Element adopted and limitation
Motamedi et al. (2024)	Platelet-demand forecasting; statistical and multivariable methods assessed with rolling-window evaluation.	**Adopt:** chronological evaluation and baseline-first comparison. **Limit:** clinical predictors and real-data accuracy do not transfer automatically to this synthetic study.
Motamedi et al. (2025)	Data-driven forecasts inform target inventory and ordering in a centralized multi-hospital setting.	**Adopt:** forecast-to-decision architecture and shortage/expiry comparison. **Limit:** pooling and operating scale differ from our simulated single centre.

Guan et al. (2017), fifth reviewed paper: Stanford forecast-to-ordering study; retrospective modelling reported 3.2% expiration versus 10.5% historically. We use Table S3 transfusion means as proxies for synthetic requests, not the raw data or Guan's model.

Synthesis: Five papers—four methodological foundations plus Guan as the calibration source—motivate chronological forecasting and testing forecasts through inventory decisions.

3. Literature Review (2/2) — Age and Simulation

Paper	Method / context	Element adopted and limitation
Haijema et al. (2007)	Dynamic programming and simulation for platelet production and replenishment.	**Adopt:** age-sensitive concepts; our simulator uses FEFO and five remaining-life cohorts. **Limit:** young/any-age eligibility is outside our pooled-demand reference.
Pauls-Worm et al. (2014)	Approximate MILP for a perishable food product with non-stationary demand, age effects and service constraints.	**Adopt:** perishable-age accounting concepts. **Limit:** food setting; its MILP is not implemented here.

Synthesis: Five papers are reviewed: four methodological foundations and Guan et al. (2017) as the calibration source. Our framework combines chronological forecasting with remaining-life accounting to compare existing approaches under common assumptions.

Core papers: Haijema et al. (2007); Pauls-Worm et al. (2014). Detailed formulations are partly unverified because authorized full texts were not supplied.

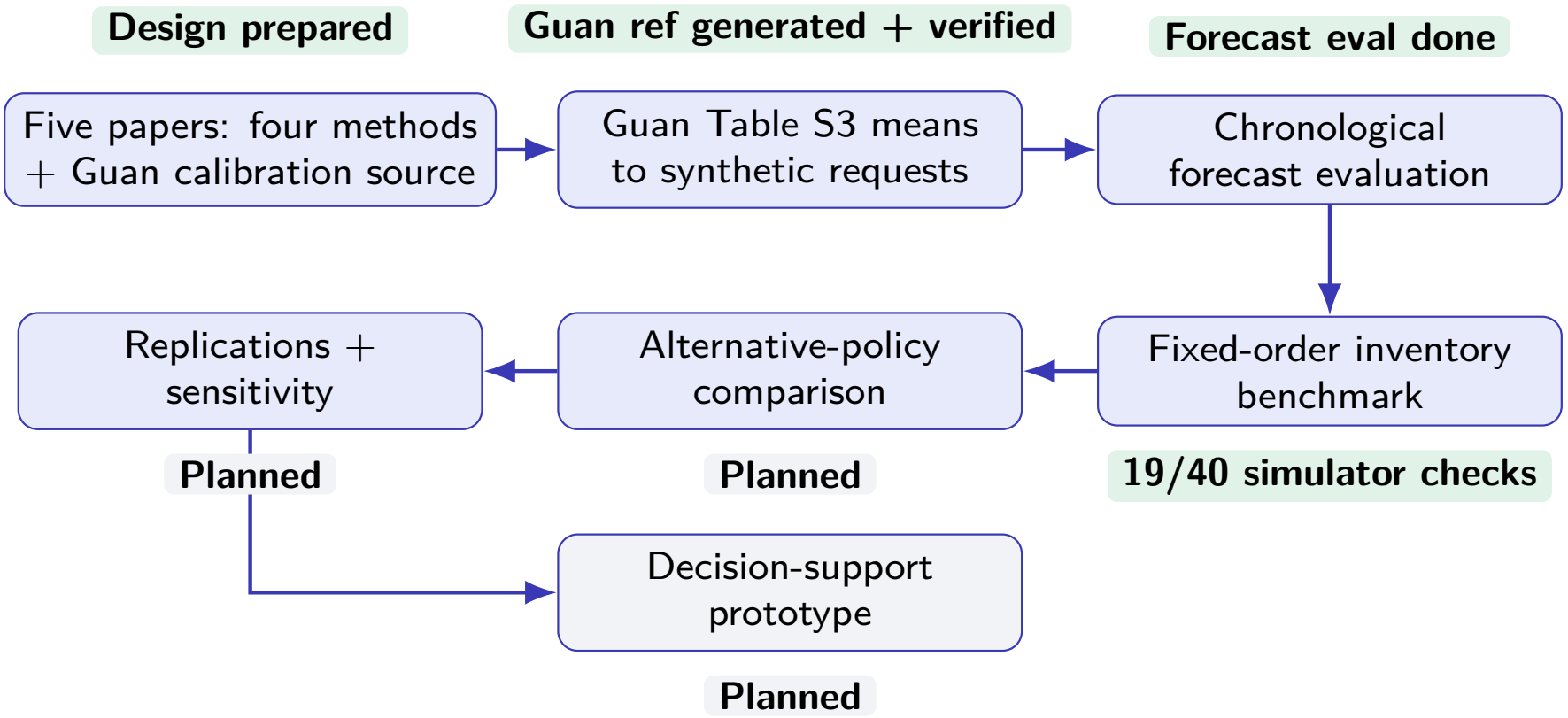
4. Research Gap and Proposed Contribution

- Guan et al. (2017) already link platelet-use prediction with collection and ordering decisions; other reviewed work informs forecasting and age-sensitive inventory. We will compare their effects under a common, disclosed single-bank simulation contract.
- Compare forecast methods within the same replenishment-rule structure to test whether lower forecast error translates into fewer unmet units or less expiry.
- Hold the forecast method fixed when comparing quantity-based and age-adjusted ordering to isolate the effect of remaining-life information.
- Test whether rankings change with demand variability, supply variability and remaining life across independent random replications.

Research question. Under the stated single-bank assumptions, how do forecast selection and remaining-life information affect platelet shortage–wastage trade-offs, and when is a simpler replenishment approach sufficient?

Contribution. A reproducible comparison of forecast selection and inventory-age adjustment on matched paths and planned replications, with recommendations conditional on evaluated assumptions.

5. Methodology — Proposed Evidence Pipeline



Proposed operating contract: one-day lead time; daily review; FEFO issuing; unmet demand recorded without automatic backlog; common random paths for policy comparison.

Five reviewed papers: four methodological foundations plus Guan as calibration source. Forty code checks include 19 simulator tests; they verify mechanics, not operational benefit.

5. Policy Alternatives and Evaluation Plan

Alternative	Status	Policy form
Fixed daily ordering	Existing benchmark	Order a fixed quantity each day.
Constant order-up-to	Planned	Order toward a fixed inventory-position target.
Forecast-dependent order-up-to	Planned	Set the order-up-to target using forecast demand.
Age-adjusted forecast-dependent variant	Planned adaptation	Adjust the forecast-informed target for age-related expiry risk.

Fair comparison. Tune eligible parameters with comparable effort on development/validation data; freeze settings and selection criteria before final testing. Use matched demand and external-supply random streams; receipt quantities may still differ when order quantities differ.

Forecast models used inside one ordering formula are variants, not separate policy families.

Report fill rate, stockout-day rate, wastage, and mean inventory. Name only the **best-performing among evaluated alternatives under stated criteria**; report shortage–wastage trade-offs rather than claiming a universal winner.

6. Synthetic Data Design & Assumptions

Required daily data dictionary

Field	Purpose
requested	Synthetic target; includes unmet requests
received	New stock arriving that day
issued	Demand fulfilled from eligible stock
unmet	Requested minus issued
expired	Units discarded after last usable day
cohorts	Closing stock by remaining usable life
pipeline	Outstanding order due next day

Assumption register

- Guan transfusion means proxy synthetic requests.
- One reference: 730 days; scale 1.0; seed 42.
- Poisson draws use seven weekday means; no surge.
- Published means are separate from team assumptions.
- Supply, receipt life, initial conditions and lead time are simulator assumptions.
- Source SDs are not matched; dispersion sensitivity is planned.

6. Synthetic Data Design — Published Weekday Reference

Published weekday platelet-transfusion means used for synthetic calibration.

Weekday	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Published mean	37.5	37.3	39.2	37.8	40.5	27.2	28.4

$$\lambda_t = \mu_{\text{weekday}(t)}, \qquad D_t \sim \text{Poisson}(\lambda_t).$$

A simulated Monday has an expected request count of 37.5 units. The generated integer count varies randomly.

Equal-weekday expected mean: **35.4143 units/day**; scale 1.0; no imposed surge.

Guan et al. (2017), PNAS, Table S3, DOI: 10.1073/pnas.1714097114. Transfusion means proxy synthetic requested demand; Poisson is a modelling assumption.

6. Synthetic Data Design — Scenario Generator

1. Assign each artificial date its weekday.
2. Read the corresponding Guan mean.
3. Draw one nonnegative integer Poisson request count.
4. Assign chronological training, validation and test labels.
5. Export requested demand without generator means or cycle features.

Guan reference | scale 1.0 | 730 days | seed 42 | surge disabled

One generated synthetic reference; not the Guan hospital dataset. Separate streams govern demand, external receipt shocks and receipt shelf lives.

7. Daily Inventory Simulation

1. Receive the order placed yesterday.
2. Review five life cohorts and place an order for tomorrow.
3. Realize requests and issue the oldest eligible stock first.
4. Record unmet requests and expire unused life-1 stock.
5. Age survivors and record closing stock.

Timing convention

Life 1 is usable today. Today's order arrives tomorrow. Demand is not backlogged.

19 correctness tests passed

7. Guan Reference: Fixed-Order Inventory Benchmark

Executed fixed-order benchmark

Test window: 1 July–30 December 2025; 183 days; Guan reference, seed 42.

Reference	Daily order	Fill rate	Stockout-day rate	Wastage	Mean closing stock
Guan reference	37 units	99.47%	1.64%	3.40%	80.67 units

Fill rate $= 6{,}565/6{,}600$.

Stockout-day rate $= 3/183$.

Wastage $= 233/(65 + 6{,}780)$.

Fixed-order outcomes only; no policy ranking

7. Demand Forecasting — Method and Evaluation

Predict requested platelet units for **today and tomorrow** before today's demand is observed.

TRAINING — 456 DAYS	**VALIDATION — 91 DAYS**	**TEST — 183 DAYS**
1 Jan 2024–31 Mar 2025	1 Apr–30 Jun 2025	1 Jul–30 Dec 2025
Fit candidate models	Select settings by same-day MAE	Evaluate frozen settings

Seasonal naïve
Use the same weekday from last week.

Seven-day moving average
Use the last seven observed requests.

ARIMA
Model past demand, changes and errors.

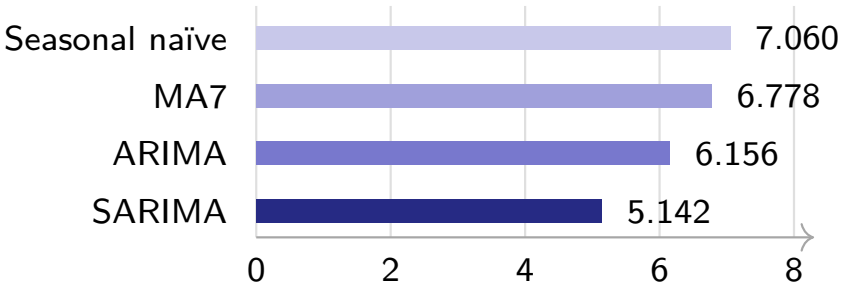
Weekly SARIMA — period 7
Represent weekly dependence; consider differencing during model selection.

Target: daily requested units, including unmet requests. Refit selected orders on all 547 development days; update with observations through yesterday. No future demand, generator means, or treatment-cycle features enter the models.

7. Forecasting Results — Model Comparison

Held-out forecast errors in synthetic requested units

Today MAE by model; zero-based axis.



MAE at both forecast horizons.

Model	Today	Tomorrow
Seasonal naïve	7.060	7.055
MA7	6.778	6.703
ARIMA	6.156	6.127
SARIMA	5.142	5.104

SARIMA had the lowest MAE and RMSE at both horizons for this single-seed reference. Today MAE reduction vs. MA7: 24.15%. Full RMSE values are in speaker notes.

7. Forecasting Integration and Next Experiment

Completed and checked

- **40 automated code tests passed**: 19 simulator, 15 integration, 6 forecasting.
- **2,192 saved forecast records audited**.
- **Two live callbacks** matched offline forecasts.
- **Six full-prefix filter checks**; inventory and request conservation passed.

Future inventory experiment

- Tune buffers using validation data only.
- Add explicit age-aware ordering.
- Compare inventory policies on identical demand and supply paths.
- Measure fill rate, shortages, expiry, and inventory.

Planned heuristic for the next experiment

$$Q_t = \text{round}_{\text{half-up}}[\max(0, F_{\text{today}} + F_{\text{tomorrow}} + b - I - P)]$$

F: point forecasts; b: validation-tuned buffer; I: available stock; P: pipeline due within the protection horizon. Age adjustment must account for expiry before future demand.

Fixed-order benchmark complete; alternative policy tests planned

8. Data and Results — Evidence Status

Evidence status

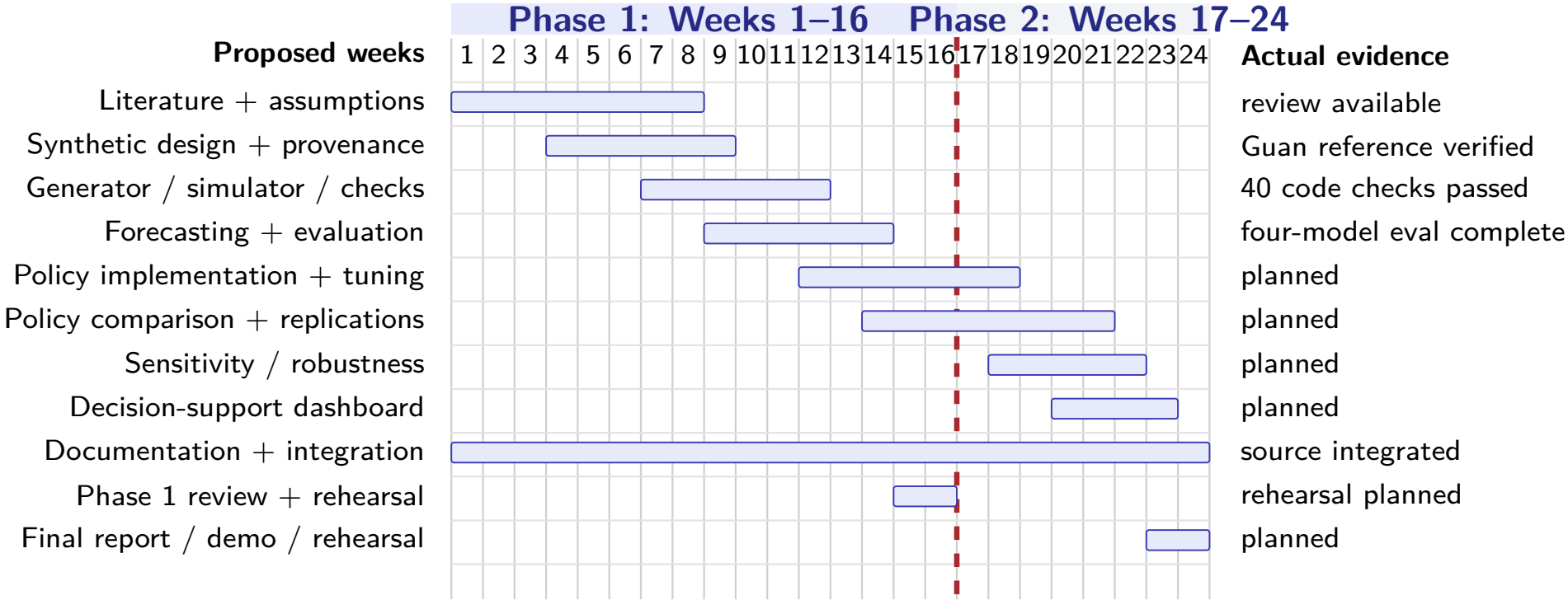
- Guan-calibrated synthetic reference, simulator checks and fixed-order benchmark: **completed**.
- Four-model forecast evaluation and callback audit: **completed**.
- SARIMA had the lowest forecast errors for this **single Guan reference and seed**.
- Alternative-policy comparison, replications, sensitivity and prototype: **planned**.

Forecast accuracy does not establish inventory improvement.

Evaluation status

Forecasting	Completed: MAE and RMSE
Service	Future policy comparison
Waste	Future policy comparison
Inventory	Future policy comparison
Validity	Planned sensitivity

8. Proposed Project Plan and Remaining Work



Phase 1: preserve Guan seed-42 reference and provenance. **Phase 2:** compare policies; start with 30 independent replications per defined scenario; assess Monte Carlo precision and sensitivity; build the prototype.

9. Conclusion

1. Executed
Published-weekday-mean reference, fixed-order benchmark and four-model forecast comparison.

2. Forecast result
Weekly SARIMA had the lowest forecast errors among four methods for this reference.

3. Policy test
Test forecast selection and remaining-life adjustment against simpler policies.

4. Robustness
Replications and sensitivity are needed before recommendations can be considered stable.

Results depend on disclosed synthetic assumptions. The current evidence does not identify the best replenishment policy or demonstrate clinical benefit.

10. References (1/2) — Main Reviewed Papers

1. Motamedi, M., Dawson, J., Li, N., & Down, D. (2025). Blood platelet inventory management: Incorporating data-driven demand forecasts. *Health Care Management Science, 28*, 191–206. doi:10.1007/s10729-025-09706-w
2. Haijema, R., van der Wal, J., & van Dijk, N. M. (2007). Blood platelet production: Optimization by dynamic programming and simulation. *Computers & Operations Research, 34*(3), 760–779. doi:10.1016/j.cor.2005.03.023
3. Motamedi, M., Dawson, J., Li, N., Down, D. G., & Heddle, N. M. (2024). Demand forecasting for platelet usage: From univariate time series to multivariable models. *PLOS ONE, 19*(4), e0297391. doi:10.1371/journal.pone.0297391
4. Pauls-Worm, K. G. J., Hendrix, E. M. T., Haijema, R., & van der Vorst, J. G. A. J. (2014). An MILP approximation for ordering perishable products with non-stationary demand and service level constraints. *International Journal of Production Economics, 157*, 133–146. doi:10.1016/j.ijpe.2014.07.020
5. **Calibration source:** Guan, L., Tian, X., Gombar, S., Zemek, A. J., Krishnan, G., Scott, R. A. H., Narasimhan, B., Tibshirani, R., & Pham, T. D. (2017). Big data modeling to predict platelet usage and minimize wastage in a tertiary care system. *Proceedings of the National Academy of Sciences, 114*(43), 11368–11373. doi:10.1073/pnas.1714097114. Table S3.

Four papers are methodological foundations; Guan is the fifth reviewed paper and separate numerical calibration source.

10. References (2/2) — Supporting Literature

1. Beliën and Forcé (2012). Supply chain management of blood products: A literature review. doi:10.1016/j.ejor.2011.05.026
2. Flint et al. (2020). Is platelet expiring out of date? A systematic review. doi:10.1016/j.tmrv.2019.08.006
3. Katsaliaki and Brailsford (2007). Using simulation to improve the blood supply chain. doi:10.1057/palgrave.jors.2602195
4. Van Dijk et al. (2009). Blood platelet production: a novel approach for practical optimization. *Transfusion, 49*(3), 411–420. doi:10.1111/j.1537-2995.2008.01996.x

Guan et al. (2017) appears in the main reviewed-paper group on the previous slide.